

The stratified K-folds cross-validation and class-balancing methods with high-performance ensemble classifiers for breast cancer classification

Mahesh T R^{a,*}, Vinoth Kumar V^b, Dhilip Kumar V^c, Oana Geman^d, Martin Margala^e,
Manisha Guduri^e

^a Department of Computer Science and Engineering, Faculty of Engineering and Technology, JAIN (Deemed-to-be University), Bangalore, 562112, India

^b School of Computer Science Engineering & Information Systems (SCORE), Vellore Institute of Technology (VIT), 632014, India

^c School of Computing, Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology, Chennai, 600062, India

^d Department of Computers, Electronics and Automation, Faculty of Electrical Engineering and Computer Science, Stefan cel Mare University of Suceava, 720229, Suceava, Romania

^e School of Computing and Informatics, University of Louisiana at Lafayette, USA

ARTICLE INFO

Handling editor: Madijd Tavana

Keywords:

Machine learning
Breast cancer
Diagnostic analytics
Ensemble method
Cross-validation
Class-balancing

ABSTRACT

Breast cancer is one of the most common causes of death among women, and early diagnosis is vital for reducing the fatality rate. This study evaluates the most widely used machine-learning breast cancer prediction and diagnosis methods. We use synthetic minority over-sampling to handle imbalanced data in the breast cancer diagnosis dataset obtained from the Wisconsin Machine Learning Repository. We use a variety of machine learning algorithms, including Logistic Regression (LR), Support Vector Machine (SVM), K-Nearest Neighbours (KNN), Classification and Regression Tree (CART), Naive Bayes (NB), and well-known ensembles methods like Majority-Voting, eXtreme Gradient Boosting algorithm (XGBoost), and Random Forest (RF) for the breast cancer classification. The findings show that the Majority-Voting ensemble method, built on the top three classifiers (LR, SVM, and CART), outperforms all other individual classifiers and offers the highest accuracy of 99.3%.

1. Introduction

One of the most prevalent and dangerous cancers in women is breast cancer, which has the potential to greatly infiltrate and spread to other tissues and organs [1,2]. This cancer was one of the most prevalent malignancies identified in 2021, with an incidence of roughly 11.7% of all new cancer cases, according to GLOBOCAN statistics (2261419 people). Moreover, 6.9% of all deaths connected to cancer were attributable to it, making it the seventh most common cause of death (684, 996) [3]. By 2020, 19.3 million new cases of cancer would have been discovered worldwide, up from an anticipated 10 million cases in 2000 [4]. According to the survey, the count of people receiving breast cancer diagnosis will be three times more when compared from 2020 to 2040. Breast cancer occurs when abnormal breast cells grow and divide.

The volume of cancer cells in the body is characterized by its staging. The size, location, and extent of the tumour, as well as whether the disease has spread to other parts of the body, all play a significant role in this decision. The primary phases of breast cancer are as follows: the

illness is not invasive in stage 1. This indicates that it has not emerged from the breast ducts. In stage 2, adjoining breast tissue has been infiltrated by the cancer cells. In stage 3, the tumour is either less than 2 cm in diameter and has spread to the lymph nodes under the arms, or it is more than 5 cm in diameter but has not gone to the lymph nodes under the arms. At this stage, tumours may or may not impact the adjacent lymph nodes and range in size from 2 to 5 cm. Stage 4 signifies the cancer's spread from its original site. In most cases, stage 3 is known to be as locally advanced breast cancer. In stage 5, the cancer has gone to the bones, liver, lungs, or brain in addition to the breast. Metastatic breast cancer is another name for stage 5 breast cancer. More than one in six fatalities are due to cancer. This emphasises the need of funding initiatives for both cancer research and cancer prevention. In this study, the efforts are made to identify the best Machine Learning(ML) model for the prediction of breast cancer. Machine learning techniques have shown promising results in medical professionals in diagnosing breast cancer based on various features extracted from medical images or patient data.

* Corresponding author.

E-mail addresses: trmahesh.1978@gmail.com, t.mahesh@jainuniversity.ac.in (M. T R), pvkumar243@gmail.com (V.K. V), vdhilipkumar@veltech.edu.in (D.K. V), oana.geman@usm.ro (O. Geman), martin.margala@louisiana.edu (M. Margala), manisha.guduri@louisiana.edu (M. Guduri).

<https://doi.org/10.1016/j.health.2023.100247>

Received 18 July 2023; Received in revised form 5 August 2023; Accepted 17 August 2023

Available online 1 September 2023

2772-4425/© 2023 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Table 1
Simulative Configurations.

Hyperparameter	Possible Value Ranges
Max Depth	None (unlimited)
Learning Rate	0.01
Subsample	0.5
Minimum Samples Split	10
Class Balancing Method	'SMOTE'
Number of Folds	10

However, there are several challenges in building accurate breast cancer classification models. One challenge is the imbalanced distribution of classes in the dataset. In breast cancer classification, the number of malignant (cancerous) cases is often significantly smaller than the number of benign (non-cancerous) cases. This class imbalance can lead to biased models that perform poorly in detecting the minority class, which is the malignant class in this case. Class balancing methods, such as oversampling the minority class or under sampling the majority class, can help address this issue and improve the overall performance of the classification models.

Another challenge is the reliable evaluation of the classification models. Simply training a model on a single dataset and evaluating its performance on the same dataset may not provide a reliable estimate of its generalization ability. Stratified k-fold cross-validation addresses this challenge by partitioning the dataset into k folds while ensuring that each fold maintains the same class distribution as the original dataset. This technique allows for a more robust evaluation of the models by training and testing them on multiple subsets of the data.

Ensemble classifiers, which combine the predictions of multiple individual classifiers, have demonstrated excellent performance in various classification tasks. By leveraging the diversity of multiple classifiers, ensemble methods can often achieve higher accuracy and better generalization compared to individual classifiers.

1.1. Research significance

The significance of using stratified k-fold cross-validation and class balancing methods with high-performance ensemble classifiers for breast cancer classification lies in their ability to address the challenges associated with imbalanced datasets and model evaluation. By properly balancing the classes and using robust evaluation techniques, these approaches can help improve the accuracy of breast cancer classification models, enabling earlier and more accurate detection of breast cancer, which is crucial for timely intervention and improved patient outcomes.

The outcomes of the proposed work are summarized as follows:

- Use synthetic minority over-sampling to handle imbalanced data in the breast cancer diagnosis dataset.
- Utilize stratified k-folds cross-validation for increasing performance evaluation.
- Compare the performance of different machine learning algorithms used for breast cancer prediction and diagnosis.
- Compare the computation Time of different machine learning algorithms to determine the optimal algorithm
- Show majority-voting ensemble technique outperforms other classifiers and offers the highest accuracy.

The research depiction is precisely delineated as follows: section 2 elaborates the significance of few most related breast cancer approaches, section 3 briefs the methodology with novel strategy and techniques, section 4 provides observed empirical outcomes and its associated discussions with detailed analysis, and lastly, section 4 presents the key-notes of the research work as notable conclusion with possible future task.

2. Related works

There are many ML techniques that can be used to predict and find breast cancer. By using a different kind of datasets, including the dataset of SEER, mammography images, Wisconsin ML Dataset, and also, dataset from other hospitals, many researchers have advanced their studies of breast cancer. Researchers might finish their work by selecting and extracting various attributes from these datasets. These are significant studies. Based on his overall findings, author Sudarshan Nayak [5] concludes that SVM is the best supervised ML algorithm for classifying breast cancer from 3D images.

Identifying breast cancer manually requires a lot of time. Even though there are still numerous diagnostic techniques accessible, proper classification requires more effort. Machine learning techniques are mostly employed in the detection of breast cancer. In contrast to other ML approaches that were employed for breast cancer identification, their work [6] compares the Relevance vector machine (RVM), which offered Low computational cost. Their research sought to compare and clarify why RVM was superior to other ML systems for detecting breast cancer malignancy, even when factors were minimised. They claimed that RVM as the best classifier as the performance secured was 97%. SVM was shown to be effective in predicting and diagnosing breast cancer by Hiba Asri [7], who also achieved the greatest results in terms of precision and low error rates, with an accuracy of 97.13%. In recent work, the authors of this study [8] proposed a similar comparison of ML algorithms as well as discovered that the SVM as the optimal classifier with 97.9% accuracy when compared with other ML techniques. A classification value of 98.4% was discovered by the author Latchoumiet TP [9], who proposed an optimization weighting of the particle swarm based on the SVM for the classification. The authors in this study [10] presented different statistical methods for breast cancer detection.

To choose the best ML method, the authors in this study, did a comparison of the effectiveness of the SVM, NB, KNN, and C4.5. SVM has proven to be the recommended classifier, with a 96.99% accuracy rate [11]. In order to figure out the optimal classifier component in breast cancer data sets, author V. Chaurasia analysed the performance criteria of supervised learning classifiers including NB, SVM-RBF kernel, RBF neural networks, Decision Tree (DT), J48, and basic classification and regression tree (CART). In the Wisconsin breast cancer data sets, the experimental results demonstrated that the classifier SVM-RBF is preferable than other ML classifiers, achieving accuracy of 96.84%. On the WBC database, the author N. Khuriwal evaluated an ensemble of two ML algorithms, Artificial Neural Networks (ANN) and LR for identifying and diagnosing breast cancer. When contrasted with relevant research from the literature and ongoing studies. It has been demonstrated that the logistic algorithm used in the ANN technique achieves 98% accuracy [12]. The authors also examined the performance based on RF and majority voting. Gradient-boosted Tree outperformed all other single and ensemble classifiers, according to performance evaluations, and achieved 90% accuracy [13]. According to the authors of this study [14], sensitivity was better with NB than ANN (100.0% versus 94.4%), but specificity was identical for both models. The ANN's accuracy was 93.3%, and the NB model's matching was 96.7%. ML classifiers have demonstrated potential for use in this study's follow-up when there was a suspicion of disease progression. The comprehensive study on breast cancer detection using ML approaches in the recent years is recorded in Table 1.

The authors in this study, have implemented optimization techniques on Wisconsin dataset providing 96.99 accuracy [15]. Saritas et al. [16] implemented ANN on the University of Coimbra dataset resulting in 86.95%. This study implemented SVM on Wisconsin providing 94.3% accuracy [17]. Vaka AR et al. [18] implemented deep neural network on M. G Cancer Hospital and Research Institute, Visakhapatnam, India with an accuracy 97.21%. This study [19] used Fuzzy ELM-RBF on Wisconsin resulting an accuracy of 98.05%. The authors [20] used K-NN on Wisconsin producing an accuracy of 95.90%. This study [21] used CNN on

some private dataset providing 98.10% accuracy. The authors in this study [22] used CNN on BreakHis Dataset with an accuracy of 95%. This study [23] used deep learning on Mammograms with 98% accuracy. Ajay et al. [24] used SVM to produce 97.13% accuracy on BCDW-11 dataset and 97.89% accuracy on WBCD-32. This study [25] uses DNN on Mammogram dataset with an accuracy of 96%. This study [26] claims that they achieved 96% area under ROC Digital Database for Screening Mammography using deep learning. This research study [27] used deep learning on Wisconsin to achieve 96.99%.

In order to discover significant chronic risk factors of novel, contagious diseases as early as feasible during pandemics, the authors in this [28] presented a hybrid Artificial Intelligence (AI) system. The suggested framework combines evolutionary search algorithms with machine learning, new Explainable AI (XAI) methods, and evolutionary search algorithms to identify the most important risk factors, use them to identify patients at high risk of mortality, and analyse the risk factors individually for each high-risk patient. While their findings point to 'age' as the main risk factor, examining the chronic risk factors specific to each age range (e.g., 18–30, 30–45, etc.) and contrasting the similarities and variances may prove to be even more illuminating. Unfortunately, the data set employed in this study did not allow them to train distinct predictive and explanatory models because there were not enough positive instances in the younger age categories.

The authors [29] used machine learning to create an AI-based disease detection system. They discussed data processing, which includes converting categorical columns and working with categorical variables. They also highlighted the key stages of application development and closely followed the software engineering process. To validate the system, they employed a random forest classification algorithm and carried out data analysis. The use of computational methods that are reliable and effective at foretelling the development of heart disease is a prerequisite for intelligent learning. For the aim of predicting and diagnosing cardiac disease, this prediction algorithm is described in this study. This classification method depends on several data cleaning approaches as well as pruning.

The authors [30] suggested Synthetic Minority Over-Sampling Technique (SMOTE) and cluster-then-predict approaches as a prediction tool for early detection of the disease. Results showed that utilising the k means approach to identify clusters within the dataset to produce another risk factor for disc degeneration enhanced the performance of the models. Performance across all five performance measures improved for all models. This lends support to earlier studies on the cluster-then-predict model technique, which uses k means to find clusters within a given data set to improve model performance. Results supported earlier studies by demonstrating an improvement in the minority class prediction when the SMOTE approach was used. However, the results demonstrated an accuracy of 94% only with bagging technique, which is comparatively less than that can be improved using majority-voting techniques.

In this study [31], the Breast Cancer Wisconsin Diagnostic dataset was subjected to five machine learning algorithms: SVM, RF, LR, DT and KNN. After obtaining the results, a performance evaluation and comparison between these various classifiers is conducted. The major goal of this study was to determine which machine learning algorithms were most efficient in terms of confusion matrices, accuracy, and precision when used to predict and diagnose breast cancer. Support vector machines were shown to perform better than all other classifiers and to have the greatest accuracy (97.2%).

Using the Wisconsin Diagnostic Breast Cancer dataset and the 10 fold cross validation technique, this paper [32] provided a thorough analysis of the classification algorithms SVM, J48, NF, and RF in terms of their prediction accuracy. Results: The SVM had the highest prediction accuracy of 97.89% and the lowest error rate of 0.14% in this investigation.

Particle Swarm Optimization (PSO) is recommended in this article [33] as a way to maximise the effectiveness of the classification with the

Decision Tree algorithm on a dataset for Wisconsin Breast Cancer. The results show that the system performed with 92.26% accuracy when compared to current best practises. By developing an early detection system for diagnosis based on a machine learning technique, the system will help reduce the prevalence of breast cancer disease and assist healthcare professionals in making decisions.

By finding local discriminating regions of a chest X-ray image by weekly supervised learning and deriving rules, and then explaining why the DL approach delivers such results, the authors simulated the decision-making process of the radiologist in this paper [34]. The study [35] demonstrates the efficacy of using data augmentation and hyper-parameter optimization techniques to enhance the performance of ML models for diagnosing breast cancer, which has important implications for the medical field where the rapid and accurate diagnosis of breast cancer is crucial with an accuracy of 99.12%. This study [36] uses a blockchain-enabled internet of medical things system for breast cancer detection. The Stacking technique was employed in this study [37] to combine three approaches with equal weights: Boosting, Bagging, and Voting. On the Wisconsin Diagnosed Diagnostic Breast Cancer (WDBC) dataset, the suggested technique produced a 97% F1-score and 98% precision. The technique was used by the authors in this work [38] on a dataset related to Parkinson's disease. The system is given the templates for the random forest and support vector machine algorithms. Random forests and support vector machines were able to produce accuracy levels of 80% and 75%, respectively. The algorithm chose the best models for each test case based on user preferences and system performance metrics. In three test cases, the support vector machine was chosen as the best model, while in one test case, the random forest was chosen as the best model.

The authors [39] used statistical features approaches and the Grey Level Co-occurrence Matrix (GLCM) to uncover underlying input patterns from skin photos. Three computationally efficient machine learning techniques, DT, SVM and KNN classifiers are applied using the extracted features for effectively classifying the skin images as melanoma (MEL), melanocytic nevus (MELNV), basal cell carcinoma (BCC), actinic keratosis (AK), benign keratosis (BK), dermatofibroma (DF), vascular lesion and Squamous cell carcinoma (SCC). ISIC 2019 Challenge and HAM10000 are two common datasets used to validate the models. SVM outperforms the other two classifiers by a small margin. This study's main goal [40] is to use several machine learning methods to anticipate the disease. Additionally, the prediction accuracy of these models is assessed together with other performance measures like precision, recall, and F1 score in order to compare them and determine which one is the most useful in this respect. With an accuracy of 82.26%, RF greatly beat the other models under investigation. Many medical problems are solved using deep learning (DL), ML, and transfer learning (TL) techniques [41]. The classification and diagnosis of cancer tumours using various types of models has been the subject of numerous scientific investigations in the past literature, however they have significant drawbacks. However, the absence of a dataset makes study difficult.

In the proposed study, the different ML algorithms namely LR, SVM, KNN, Classification and regression tree, NB, and well-known ensembles methods like Majority Voting, eXtreme gradient boosting algorithm as well as RF employed on the publicly available UCI ML repository. SMOTE technique is used to handle the dataset as it is imbalanced. The study recommends the best ML model for the prediction of breast cancer. The remaining portion of this study is structured as follows: The next section 3 presents various ML algorithms used for classification of the breast cancer. Section 4 details the evaluation of all the ML algorithms against various performance metrics and recommends the best model. Finally, the last section concludes with the recommendation of the best ML model.

3. Methodology

Data gathering is the first phase in our methodology, which is

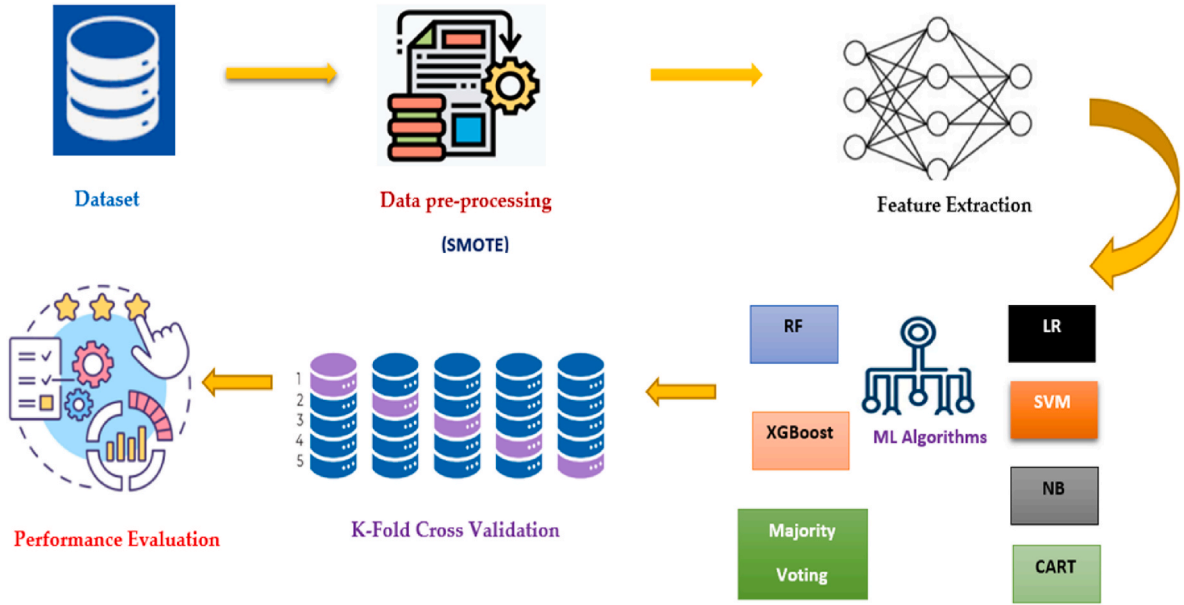


Fig. 1. Operational Flow of proposed breast cancer classification.

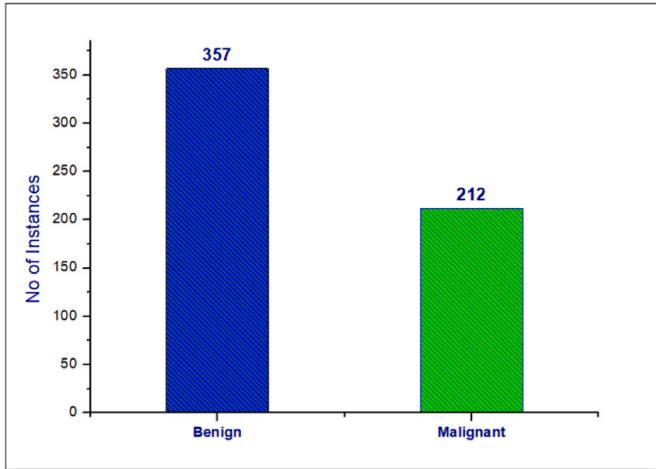


Fig. 2. Malignant and Benign instances of the dataset.

followed by pre-processing, which entails four steps: data cleaning and editing, attribute selection, setting the target Role, and feature extraction [42]. Using the 10-fold cross-validation strategy, the research provides a solution to the question of selecting the algorithm that offers the highest level of accuracy and best utilises efficacy and efficiency for the identification of breast cancer. Fig. 1 depicts the recommended diagram.

We have used the publicly available Wisconsin dataset from UCI ML repository [43]. There are total 569 instances in the date set. There are 32 attributes. The dataset contains 357 benign, 212 malignant instances resulting in 37.26% Malignant and in 62.74% Benign as shown in Fig. 2. This clearly shows that there is an imbalance in the dataset. So as to handle this imbalance, SMOTE technique is used.

The mean approach was used to calculate missing values that are numeric, while the mode has been utilized to calculate missing values that are nominal during pre-processing. RFE which is a Recursive Feature Elimination approach has been utilized to select the aspects of relevance related to the attributes of high importance for the diagnosis of breast cancer [44–46].

The effectiveness of SMOTE depends on its capacity to produce

patterns that closely resemble the underlying distribution [47,48]. The two key variables that define every distribution are the mean vector as well as the covariance matrix. For the SMOTE-generated patterns' mean and covariance matrix, we create approximations of the formulas and compare them to the actual results as shown in equation (1).

$$Z = X_0 + w\Delta \quad (1)$$

Let $\Delta = X - X_0$ then:

Where w denotes generated number in $[0, w^*]$. w^* is either more than or equal to one and allows us to interpolate on the line connecting the pattern X_0 as well as its randomly chosen neighbour X .

$$E[Z] \approx \mu_{X_0} + \frac{Cw^*}{2} \int_{X_0} p(X_0)^{-\frac{2}{d}} \frac{\partial p(X_0)}{\partial X} dX_0 \quad (2)$$

The final approximations of the mean and covariance matrix of the generated pattern vector Z are given by the following Eq. (2) and Eq. (4) respectively. ∂X is computed as shown in equation (3).

$$\text{Where } \partial X \left[\frac{\partial p(X)}{\partial X} \right]^T = \left(\frac{\partial p(X)}{\partial X_1}, \dots, \frac{\partial p(X)}{\partial X_d} \right) \quad (3)$$

$$\begin{aligned} \sum_Z &= \sum_{X_0} + \frac{Cw^{*2}}{3} \int_{X_0} p(X_0)^{1-\frac{2}{d}} dX_0 I \\ &\quad - \frac{C^2w^{*2}}{3} \int_{X_0} p(X_0)^{1-\frac{2}{d}} \frac{\partial p(X_0)}{\partial X} dX_0 \int_{X_0} p(X_0)^{-\frac{2}{d}} \frac{\partial p(X_0)}{\partial X} dX_0 \\ &\quad + \frac{Cw^{*2}}{2} \left[\int_{X_0} p(X_0)^{-\frac{2}{d}} \frac{\partial p(X_0)}{\partial X} [(X_0 - \mu_{X_0})^T] dX_0 \int_{X_0} p(X_0)^{-\frac{2}{d}} (X_0 - \mu_{X_0}) \frac{\partial p(X_0)}{\partial X} dX_0 \right] \end{aligned} \quad (4)$$

$$C = \frac{N! \Gamma(1 + \frac{2}{d}) \Gamma(K + \frac{2}{d} + 1)}{\pi K! (d + 2) \Gamma(N + \frac{2}{d} + 1)} \quad (5)$$

Where d denotes the pattern vector dimension, n represents the number of original patterns, μ_{X_0} denotes the true mean vector, $\sum_{X_0}$ denotes true covariance function, $p(X_0)$ is the class conditional density, I is the identity matrix and C is computed as (4) [49].

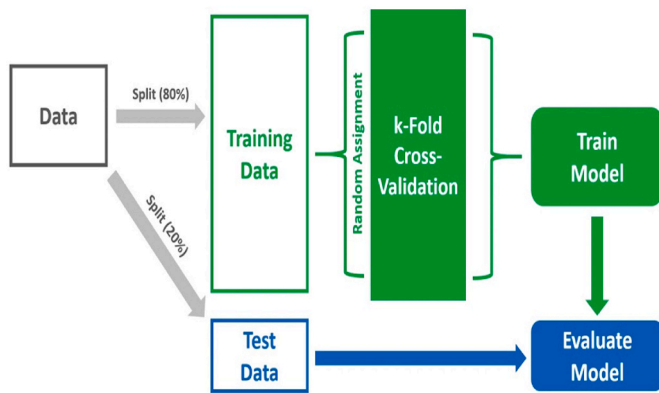


Fig. 3. K-fold cross validation.

After the data pre-processing technique, the data set is split into 80% of the records called training data and 20% of the data called test data. Then the different classifiers LR, SVM, KNN, NB, CART and popular ensembles methods like Majority-Voting, XGBoost classifier and RF are implemented on the publicly available dataset. The procedure of K-fold cross approach is shown in Fig. 3.

In K-fold cross validation, the population is randomly split into k folds or groups ($k=10$ in the proposed study) that are roughly similar in size. The technique is then fitted on the remaining $k-1$ folds using the initial fold as a validation set. The general procedure is shown in Algorithm1.

Algorithm 1. 10-fold cross validation procedure.

1. Shuffle the dataset randomly.
2. Split the dataset into k groups
3. For each unique group:
 - a. Take the group as a hold out or test data set
 - b. Take the remaining groups as a training data set
 - c. Fit a model on the training set and evaluate it on the test set
 - d. Retain the evaluation score and discard the model
4. Summarize the skill of the model using the sample of model evaluation scores

The EnsembleVoteClassifier is a meta-classifier used to combine machine learning classifiers that are conceptually similar or dissimilar for classification using majority or plurality vote. In hard voting, we identify the final class label as the one that the classification models have predicted most frequently.

In voting classifiers there are two classifications: hard voting classifier and soft voting classifier. In hard voting classifier, if maximum number of models are giving the output as "Breast Cancer Disease", then, it predicts for the test data, the prediction as "Breast Cancer Disease". This technique is very simple and this is how the voting classifier works and getting used in ensemble technique as well. In case of soft voting classifier, the probability of every model in predicting the disease or not is computed. The mean values of prediction are compared, whichever is more that becomes the outcome of the voting classifier. In the next section, the performance of all ML algorithms are evaluated against different metrics. Among all ML techniques, we identify the top three performers which are providing good precision, sensitivity and f1 score with best time taken to construct the model as base classifiers. Then the majority voting ensemble classifier is built with these top three classifiers and then the performance of this is evaluated and compared with other ML models.

Table 2
Accuracy of classifiers.

Evaluation Criteria	LR	SVM	KNN	NB	CART	RF	XGBoost	Majority Voting
Time to construct model(seconds)	0.04	0.03	0.14	0.10	0.20	0.21	0.22	0.11
Correctly classified Instances (%)	98.891	98.723	96.923	98.772	96.124	97.192	97.922	99.312
Incorrectly classified Instances (%)	1.109	1.277	3.077	1.228	3.876	2.808	2.078	0.688
Accuracy (%)	98.8	98.7	96.9	98.7	96.1	97.2	97.9	99.3

4. Results and discussions

4.1. Experimental layout

The scikit-learn v0.24.2 library function is incorporated in python v3.9 for the implementation of chosen ML algorithms. In addition to this, we included caret v6.0 from R language for class balancing (upsampling), and statistical computing. Table 1 represents the simulative configuration of the research work.

4.2. Outcome analysis

After using ML algorithms on the Wisconsin Diagnostic dataset for breast cancer, we evaluated and compared the models using Confusion time to create the model (in seconds), Precision, Accuracy, Sensitivity, F1 Score, Matthew's Correlation coefficient (MCC) and Area under curve (AUC) as performance measures, and then top algorithm for predicting

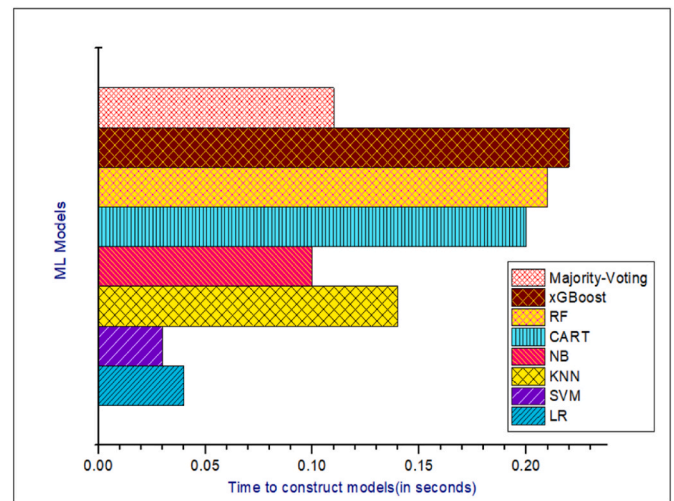


Fig. 4. Time to construct models(in seconds).

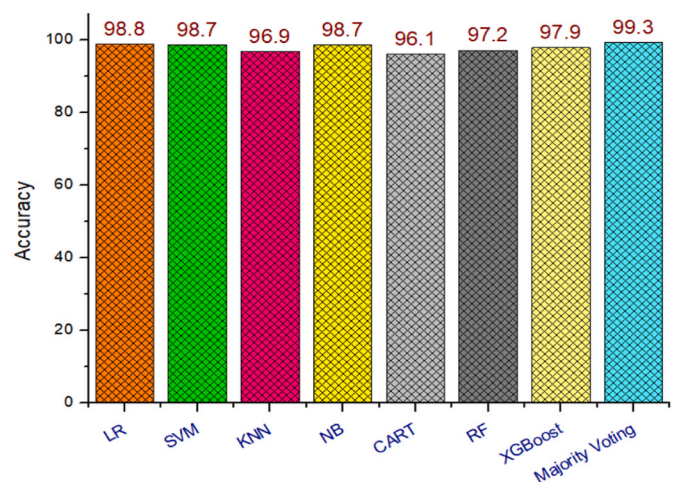
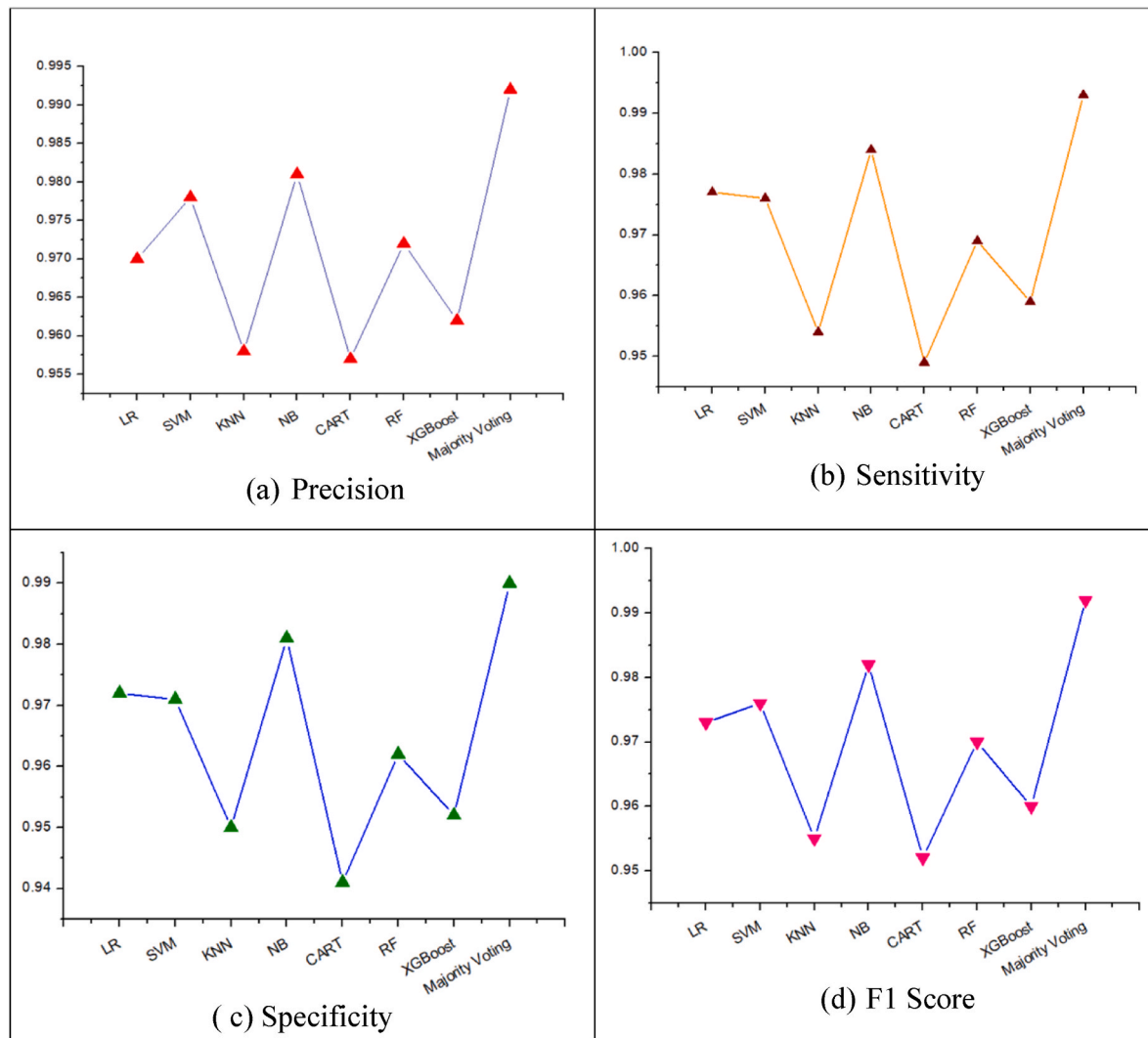


Fig. 5. Accuracy of ML models.

Table 3

Performance of ML classifiers.

Evaluation Criteria	LR	SVM	KNN	NB	CART	RF	XGBoost	Majority Voting
Precision(%)	0.97	0.978	0.958	0.981	0.957	0.972	0.962	0.992
Sensitivity(%)	0.977	0.976	0.954	0.984	0.949	0.969	0.959	0.993
Specificity(%)	0.972	0.971	0.95	0.981	0.941	0.962	0.952	0.99
ROC Area	0.971	0.975	0.955	0.986	0.956	0.975	0.964	0.995
F1 Score(%)	0.973	0.976	0.955	0.982	0.952	0.97	0.96	0.992

**Fig. 6.** Performance of different classifiers.**Table 4**

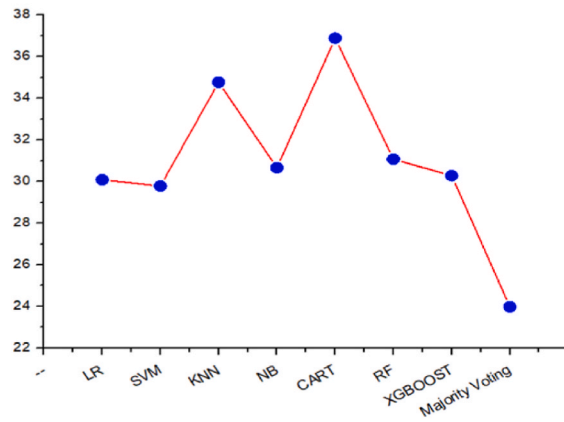
Simulation and training errors.

Evaluation Criteria	LR	SVM	KNN	NB	CART	RF	XGBOOST	Majority Voting
Kappa	0.967	0.957	0.934	0.976	0.956	0.78	0.971	0.991
MAE	0.023	0.021	0.123	0.032	0.102	0.032	0.033	0.0192
RMSE	0.182	0.190	0.199	0.189	0.178	0.156	0.151	0.141
MCC	0.98	0.97	0.96	0.98	0.95	0.96	0.97	0.99
MAPE	0.032	0.029	0.034	0.030	0.035	0.037	0.030	0.024
RAE(%)	3.982	4.092	5.345	4.044	5.012	5.892	4.012	3.134
RRSE(%)	30.08	29.78	34.77	30.66	36.88	31.07	30.28	23.98

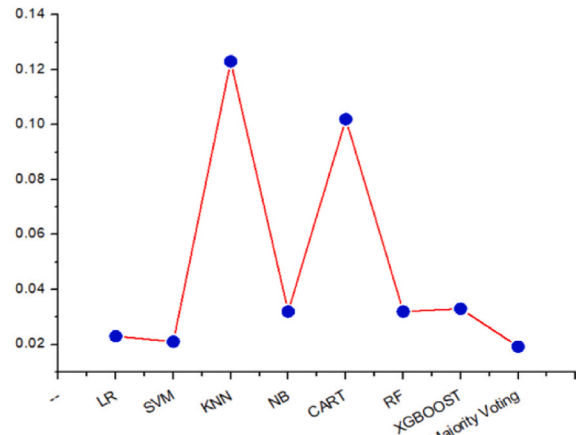
breast cancer is chosen. The parameters derived from the confusion matrix are True Positive Score (TPS), True Negative Score (TNS), False positive Score (FPS) and False Negative Score (FNS). Out of all the positive forecasts, precision tells us how many are truly accurate. When

False Positives are more problematic than False Negatives, precision is a valuable indicator. Precision is computed using (6).

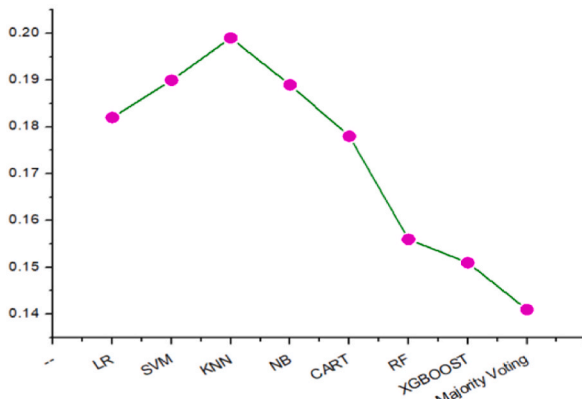
$$\text{Precision} = \frac{TPS}{TPS + FPS} \quad (6)$$



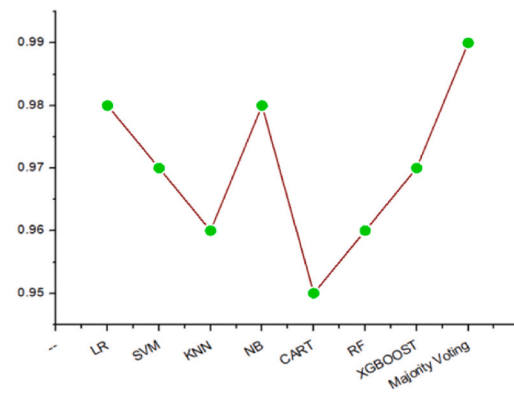
(a) RRSE



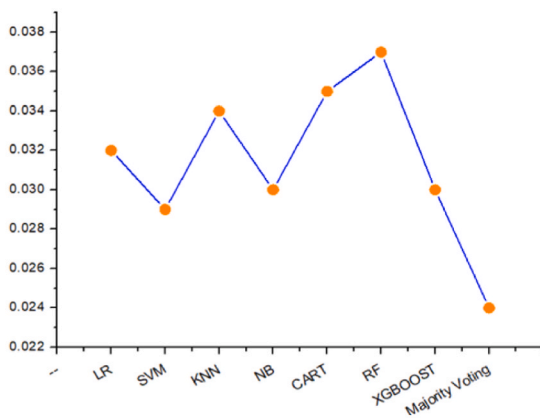
(b) MAE



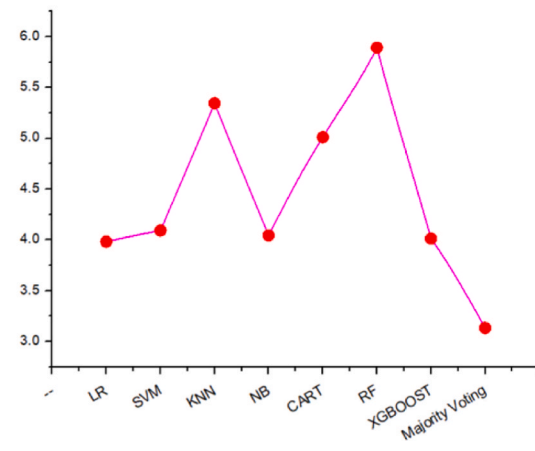
(c) RMSE



(d) MCC



(e) MAPE



(f) RAE

Fig. 7. Training and simulation error.

Sensitivity (SN) is also referred as recall. When False Negative beats False Positive, recall is a meaningful statistic. It is computed using (7).

$$SN = \frac{TPS}{TPS + FNS} \quad (7)$$

The percentage of actual negatives that were anticipated as negatives is known as specificity (SP). Specificity plus false positive rate would always add up to one. Specificity is a measurement of the percentage of non-breast cancer patients who were correctly identified as those who

do not have the disease. In other words, the specificity of the prediction that the individual is healthy actually came true. It is computed using equation (8).

$$SP = \frac{TNS}{TNS + FPS} \quad (8)$$

The classifier's accuracy is simply measured by how frequently it predicts correctly. It is the proportion between the number of accurate forecasts and all predictions combined. Unbalanced classes are not a

Table 5
Comparison of MAE and RMSE of different algorithms.

Approaches	RMSE		MAE	
	A- RMSE	Best-RMSE	A-MAE	Best MAE
LR	0.1810	0.1792	0.0229	0.0221
SVM	0.1899	0.1892	0.0209	0.0201
KNN	0.1990	0.9880	0.1228	0.1219
NB	0.1887	0.1825	0.0319	0.0316
CART	0.1734	0.1698	0.1010	0.0992
RF	0.1552	0.1516	0.0324	0.0318
XGBoost	0.1523	0.1493	0.0298	0.0303
Majority-Voting	0.1402	0.1392	0.0934	0.0190

good fit for the accuracy metric. Accuracy is computed using the equation (9).

$$Accuracy = \frac{TPS + TNS}{TNS + TPS + FNS + FPS} \quad (9)$$

Matthew's Correlation Coefficient (MCC), is computed using equation (10). MCC is a metric used to assess the effectiveness of a binary classifier for identifying breast cancer in a patient.

$$MCC = \frac{TPS * TNS * FPS * FNS}{\sqrt{(TPS + FPS)(TPS + FNS)(TNS + FPS)(TNS + FNS)}} \quad (10)$$

The kappa statistic (KS) measures the difference between actual and observed classes after correction. Kappa is computed using the equation (11).

$$K = \frac{p_0 - p_e}{1 - p_e} \quad (11)$$

If there is a correct or proper agreement, then k value will be 1 else it will be ≤ 0 .

The difference between the real value and naive model value is measured as the relative absolute error (RAE) and it is computed using (12).

$$RAE = \frac{\sqrt{\sum_{i=1}^n (A - B)^2}}{\sqrt{\sum_{i=1}^n B_i^2}} \quad (12)$$

where A denotes actual values and B denotes the expected values. Ideal RAE values are lesser than 1.

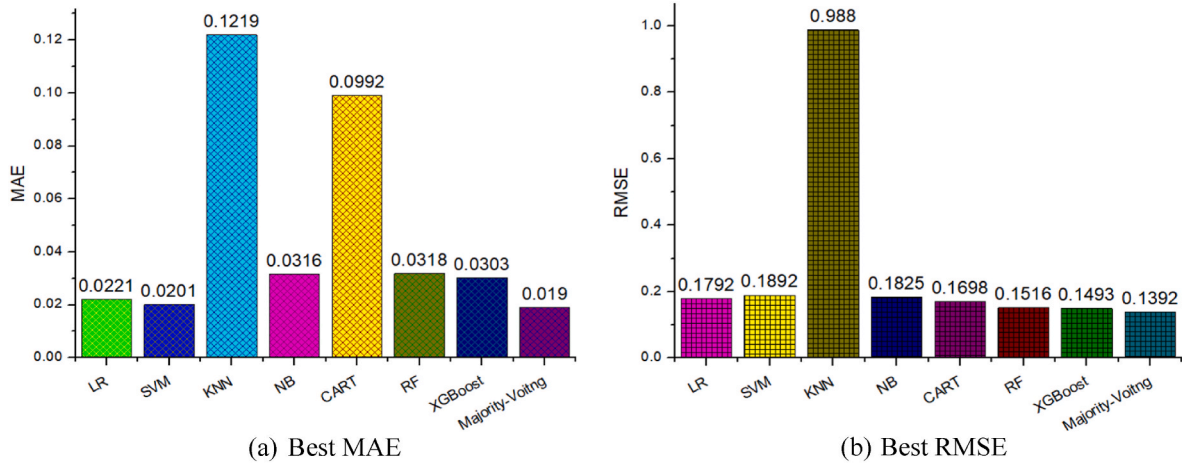


Fig. 8. Best MAE and RMSE

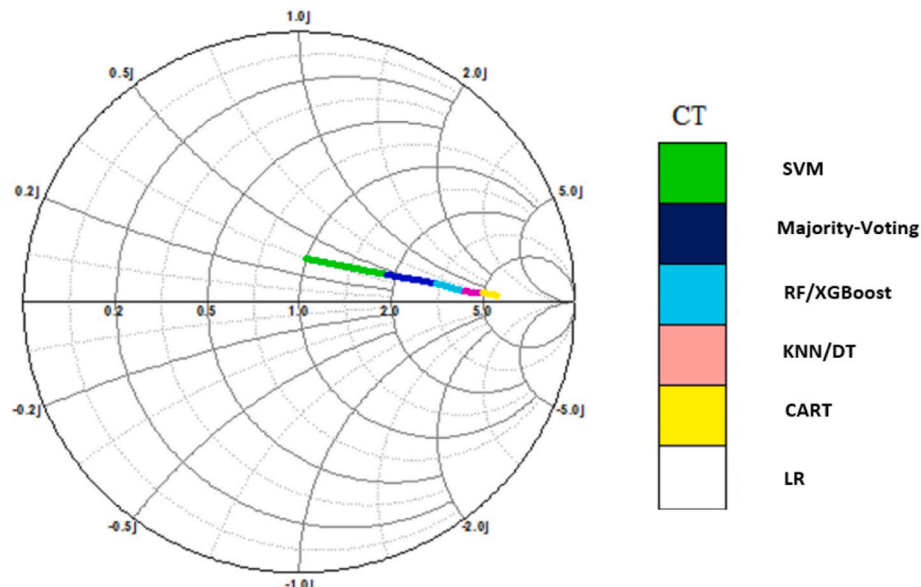


Fig. 9. Computation Time of different ML algorithms.

The average absolute error between actual and anticipated values is known as mean absolute error (MAE) and MAE is computed using (13)

$$MAE = \frac{\sum |actual - prediction|}{Number\ of\ Observations} \quad (13)$$

The mean squared difference between the expected and actual values is known as the "root mean squared error" (RMSE) and RMSE is computed using (14). The acronym MAPE stands for Mean Absolute Percentage Error (MAPE). With an increase in mistake, it rises linearly. The model's performance improves with decreasing MAPE and MAPE is computed using (15).

$$RMSE = \sqrt{\frac{\sum (actual - prediction)^2}{Number\ of\ observations}} \quad (14)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \frac{|y_i^{real} - y_i^{pred}|}{y_i^{real}} * 100\% \quad (15)$$

$$E_i = \sqrt{\frac{\sum_{j=1}^n (P_{(ij)} - T_j)^2}{\sum_{j=1}^n (T_j - \bar{T})^2}} \quad (16)$$

The error is reduced to the same dimensions as the prediction using the root relative squared error (RRSE). Total square error divided by total square error of a straightforward predictor yields relative square error. It might be between 0 and 1. A realistic model should have a value that is near to zero, while one with a number higher than 1 is not. The formula shown in (16) determines the root of the individual model j 's relative square error E_i and T is computed using (17).

$$\text{Where } \bar{T} = \frac{1}{n} \sum_{j=1}^n T_j \quad (17)$$

The performance of base classifiers LR, SVM, KNN, NB, CART and the ensemble classifiers RF and XGBoost are evaluated in this section. The effectiveness of Majority voting-based ensembles « NB, SVM and LR » is also evaluated in terms of the time taken to construct the model, number of correct predictions and incorrect predictions. These values are recorded in Table 2.

As depicted in Fig. 4, SVM has taken the least time 0.03 s to build the model, LR has taken 0.03 s however, the majority voting ensemble classifier has taken 0.11 s. It is observed that XGBoost has taken more time of 0.22 s to build the model.

As it is observed that from Table 2, the majority voting ensemble classifier has the highest accuracy of 99.3% and the accuracy of LR, SVM, KNN, NB, CART, RF and XGBoost are recorded as 98.8%, 98.7%, 96.9%, 98.7, 96.1%, 97.2%, 97.9% and 99.3% respectively. The accuracy of different models is depicted in Fig. 5.

The ML models are evaluated against different parameters like sensitivity, precision, specificity, ROC curve and F1 score. These values are recorded in Table 3.

The precision, sensitivity, specificity, ROC curve values and F1 score for majority voting-ensemble classifier are found to be 0.992, 0.993, 0.990, 0.995 and 0.992 respectively as shown in Fig. 6. KNN is the least performer with values of 0.958, 0.954, 0.950, 0.955 and 0.955 for precision, sensitivity, specificity, ROC curve values and F1 score respectively. Majority voting-ensemble classifier outperforms compared to other ML models.

The error of simulation is also considered in this study to improve the measurement of classifier performance. To do this, we evaluate the performance of our classifier using the metrics presented in Table 4: Kappa, MAE, RAE, RMSE, MCC, MAPE, and RRSE.

The error rates of different ML model along with the majority voting-ensemble technique is shown in Fig. 7. The error rates of kappa, MAE, RMSE, MCC, MAPE, RAE and RRSE for majority voting ensemble

classifier are 0.991, 0.0192, 0.141, 0.99, 0.024, 3.134, and 23.98 respectively. The majority voting ensemble classifier outperforms with least error rates compared to all other models.

The summary of the complete findings with regard to the best and average RMSE and MAE values is shown in Table 5. While Best-MAE is the minimum MAE and Best-RMSE is the least RMSE value acquired by varying the number "K" of cross validation process. A-MAE is the average of the MAE and A-RMSE is the average of the RMSE values.

The Best-MAE and Best RMSE of all the ML models is shown in Fig. 8.

From Fig. 8, it is observed that Majority-voting has the Best MAE value of 0.0190 and Best RMSE value of 0.1392. KNN has exhibited with poor performance.

Fig. 9 illustrates the processing time (Computation Time –CT) required for several classification techniques. In comparison to other methods. Majority-Voting takes longer time it is comprised of multiple individual classifiers. CG (conjugate gradient) optimization was utilized for primary LR with compliance. L2 regularization and Alpha = 0.0001 were being used by SVM to get the best learning rate. Using Gini impurity, a DT in RF was divided into smaller trees which consume optimal time for processing. It was enlarged until all leaves were pure or included fewer than two nodes, whichever came first.

5. Conclusion

Our study is primarily focused on the development of predictive models to achieve high accuracy in anticipating real disease findings using supervised ML techniques. It is possible to stop the spread of breast cancer by identifying it early and classifying it. The utilisation of data mining technologies in the medical field is crucial because they undoubtedly facilitate decision-making. Yet, to achieve this, such algorithms necessitate high performance, remarkable precision, and an excellent technique selection based on the working environment and the data being processed. In this study, the different ML models were employed on the publicly available Wisconsin ML repository having 569 records. The majority voting ensemble classifier of three classifiers (LR, SVM and NB) outperforms other models with an accuracy of 99.3%, followed by XGBoost and the Random Forest classifiers with an accuracy of 97.9% and 97.2% respectively. The majority voting ensemble classifier also results in Best RMSE and Best MAE values of 0.1392 and 0.0190 respectively.

Future studies may find it beneficially challenging to implement this framework on larger datasets and, if possible, to evaluate it on a wider scale. Additionally, it can be combined with several optimization methods, like PSO, ACO, and GA (Genetic Algorithm). These methods can be applied to precisely select the optimum ensemble algorithm settings.

Conflicts of interest

The authors declare no conflict of interest.

Funding statement

This work has not received any funds.

Declaration of competing interest

The authors declare that no conflict of interest.

Data availability

Data will be made available on request.

References

- [1] N. Pasha, N.C. Turner, Understanding and overcoming tumor heterogeneity in metastatic breast cancer treatment, *Nat. Can. (Que.)* 2 (7) (2021) 680–692.
- [2] S. Lei, R. Zheng, S. Zhang, R. Chen, S. Wang, K. Sun, et al., Breast cancer incidence and mortality in women in China: temporal trends and projections to 2030, *Cancer Biol. Med.* 18 (3) (2021) 900.
- [3] A. Kamboj, P. Tanay, A. Sinha, et al., *Breast Cancer Detection Using Supervised Machine Learning: a Comparative Analysis*, Springer, Singapore, 2021, p. 263.
- [4] WHO | Breast Cancer', WHO. <http://www.who.int/cancer/prevention/diagnosis-screening/breast-cancer/en>.
- [5] S. Nayak, D. Gope, "Comparison of Supervised Learning Algorithms for RF-Based Breast Cancer Detection," 2017 Computing and Electromagnetics International Workshop, (CEM), Barcelona, 2017.
- [6] B.M. Gayathri, C.P. Sumathi, Comparative Study of Relevance Vector Machine with Various Machine Learning Techniques Used for Detecting Breast Cancer, 2016 IEEE International Conference on Computational Intelligence and Computing Research (ICICR), Chennai, 2016, pp. 1–5.
- [7] H. Asri, H. Mousannif, H.A. Moatassime, T. Noel, Using machine learning algorithms for breast cancer risk prediction and diagnosis, *Procedia Comput. Sci.* 83 (2016) 1064–1069, <https://doi.org/10.1016/j.procs.2016.04.224>.
- [8] Y. Khoudfi, M. Bahaj, Applying Best Machine Learning Algorithms for Breast Cancer Prediction and Classification, *IEEE*, 2018, p. 978, 1-5386- 4225- 2/18/ \$31.00 ©.
- [9] L. Latchoumi, T. P. Parthiban, Abnormality detection using weighed particle swarm optimization and smooth support vector machine, *Biomed. Res.* 28 (11) (2017) 4749–4751.
- [10] S. Aruna, D. S. P. Rajagopalan, and L. V. Nandakishore, 'KNOWLEDGE BASED ANALYSIS OF VARIOUS STATISTICAL TOOLS IN DETECTING BREAST CANCER', *Computer Science*, p. 9.
- [11] S. T. Ahmed, V. Kumar and J. Kim, "AITel: eHealth augmented intelligence based telemedicine resource recommendation framework for IoT devices in smart cities," in *IEEE Internet of Things Journal*, doi: 10.1109/JIOT.2023.3243784.
- [12] N. Khuriwal, N. Mishra, Breast cancer diagnosis using adaptive voting ensemble machine learning algorithm, in: 2018 IEEMA Engineer Infinite Conference, eTechNXT), New Delhi, Mar, 2018, pp. 1–5, <https://doi.org/10.1109/ETECHNXT.2018.8385355>.
- [13] M.I. Faisal, S. Bashir, Z.S. Khan, F. Hassan Khan, in: 'An Evaluation of Machine Learning Classifiers and Ensembles for Early Stage Prediction of Lung Cancer', 2018 3rd International Conference on Emerging Trends in Engineering, Sciences and Technology (ICEEST), Karachi, Pakistan, Dec. 2018, pp. 1–4, <https://doi.org/10.1109/ICEEST.2018.8643311>.
- [14] H. Wen, S. Li, W. Li, J. Li, C. Yin, in: 'Comparison of Four Machine Learning Techniques for the Prediction of Prostate Cancer Survivability', 2018 15th International Computer Conference on Wavelet Active Media Technology and Information Processing (ICCWAMTIP), Chengdu, China, Dec. 2018, pp. 112–116, <https://doi.org/10.1109/ICCWAMTIP.2018.8632577>.
- [15] E.A. Bayrak, P. Kirci, T. Ensari, Scientific Meeting on Electrical/Electronics and Biomedical Engineering and Computer Science, vol. 2019, EBBT, 2019, <https://doi.org/10.1109/EBBT.2019.8741990>.
- [16] M.M. Saritas, A. Yasar, International journal of intelligent systems and applications in engineering performance analysis of ann and naive bayes classification algorithm for data classification, *Orig Res Pap Int J Intell Syst Appl Eng IJISAE* 7 (2019) 88–91, <https://doi.org/10.1039/b000000x>.
- [17] M.A. Rufai, A.S. Muhammad, S. Garba, et al., Machine learning model for breast cancer detection, *FUDMA J Sci* 4 (1) (2020) 55–61.
- [18] A.R. Vaka, B. Soni, SRK, Breast cancer detection by leveraging machine learning, *ICT Exp* 6 (2020) 320–324, <https://doi.org/10.1016/j.ict.2020.04.009>.
- [19] S. Mojriani, RIVF International Conference on Computing and Communication Technologies, vol. 2020, RIVF, IEEE, 2020.
- [20] S. Sharma, A. Aggarwal, T. Choudhury, Proceedings of the International Conference on Computational Techniques, Electronics and Mechanical Systems, CTEMS, 2018, pp. 114–118, <https://doi.org/10.1109/CTEMS.2018.8769187>.
- [21] S. Arooj, Atta-ur-Rahman, M. Zubair, M.F. Khan, K. Alissa, M.A. Khan, A. Mosavi, Breast cancer detection and classification empowered with transfer learning, *Front. Public Health* 10 (2022), 924432, <https://doi.org/10.3389/fpubh.2022.924432>.
- [22] Majid Nawaz, Adel A. Sewissy, Taysir Hassan A. Soliman, Multi-class breast cancer classification using deep learning convolution neural network, *Int. J. Adv. Comput. Sci. Appl.* 9 (6) (2018) 316–322.
- [23] Naresh Khuriwal, Nidhi Mishra, Breast cancer detection from histopathological images using deep learning, 3rd International Conference and Workshops on Recent Advances and Innovations in Engineering (November 2018) 22–25.
- [24] Ajay Kumar, R. Sushil, A.K. Tiwari, "Comparative study of classification techniques for breast cancer diagnosis," *Int. J. Comput. Sci. Eng.* 7 (1) (2019) 234–240.
- [25] R. Chithrakannan, P. Kavitha, T. Mangayarkarasi, R. Karthikeyan, Breast cancer detection using machine learning, *Int. J. Innovative Technol. Explor. Eng.* 8 (11) (2019) 3123–3126.
- [26] Weal E. Fathy, Amr S. Ghoneim, A deep learning approach for breast cancer mass detection, *Int. J. Adv. Comput. Sci. Appl.* 10 (1) (2019) 175–182.
- [27] Panuwat Mekha, Nutnicha Teeyasuksaet, Deep Learning Algorithms for Predicting Breast Cancer Based on Tumor Cells," 4th International Conference on Digital Arts, Media and Technology and 2nd ECTI Northern Section Conference on Electrical, Electronics, Computer and Telecommunication Engineering, 2019, pp. 343–346.
- [28] Behrooz Davazdahemami, Hamed M. Zolbanin, Dursun Delen, An explanatory analytics framework for early detection of chronic risk factors in pandemics, *Healthcare Analytics*, Volume2 (2022), 100020, <https://doi.org/10.1016/j.health.2022.100020>. ISSN2772-4425.
- [29] Victor Chang, Vallabhanent Rupa Bhavani, Ariel Qianwen Xu, M.A. Hossain, An artificial intelligence model for heart disease detection using machine learning algorithms, *Healthcare Analytics* 2 (2022), 100016, <https://doi.org/10.1016/j.health.2022.100016>. ISSN 2772-4425.
- [30] Janae Bradley, Suchithra Rajendran, Developing predictive models for early detection of intervertebral disc degeneration risk, *Healthcare Analytics* 2 (2022), 100054, <https://doi.org/10.1016/j.health.2022.100054>. ISSN 2772-4425.
- [31] Mohammed Amine Naji, Sanaa El Filali, Kawtar Aarika, EL Habib Benlahmar, Rachida Ait Abdelouahid, Debauche Olivier, Machine learning algorithms for breast cancer prediction and diagnosis, *Procedia Comput. Sci.* 191 (2021) 487–492, <https://doi.org/10.1016/j.procs.2021.07.062>. ISSN 1877-0509.
- [32] Somil * Jain, Puneet Kumar, Prediction of breast cancer using machine learning, *Recent Advances in Computer Science and Communications* 13 (5) (2020), <https://doi.org/10.2174/2213275912666190617160834>.
- [33] J.O. Afolayan, M.O. Adebisi, M.O. Arowolo, C. Chakraborty, A.A. Adebisi, Breast cancer detection using particle swarm optimization and decision tree machine learning technique, in: C. Chakraborty, M.R. Khosravi (Eds.), *Intelligent Healthcare*, Springer, Singapore, 2022, https://doi.org/10.1007/978-981-16-8150-9_4.
- [34] Shiva prasad Koyyada, Thipendra P. Singh, An explainable artificial intelligence model for identifying local indicators and detecting lung disease from chest X-ray images, *Healthcare Analytics* 4 (2023), 100206, <https://doi.org/10.1016/j.health.2023.100206>. ISSN 2772-4425.
- [35] Temidayo Oluwatolase Omothinwa, David Opeoluwa Oyewola, Emmanuel Gbenga Dada, A light gradient-boosting machine algorithm with tree-structured parzen estimator for breast cancer diagnosis, *Healthcare Analytics* 4 (2023), 100218, <https://doi.org/10.1016/j.health.2023.100218>. ISSN 2772-4425.
- [36] Sushovan Chaudhury, Kartik Sau, A blockchain-enabled internet of medical things system for breast cancer detection in healthcare, *Healthcare Analytics* 4 (2023), 100221, <https://doi.org/10.1016/j.health.2023.100221>. ISSN 2772-4425.
- [37] Mina Samieinasab, S. Ahmad Torabzadeh, Arman Behnam, Amir Aghsami, Fariborz Jolai, Meta-Health Stack: a new approach for breast cancer prediction, *Healthcare Analytics* 2 (2022), 100010, <https://doi.org/10.1016/j.health.2021.100010>. ISSN 2772-4425.
- [38] Yashodhan Ketkar, Susupti Gawade, A Decision Support System for Selecting the Most Suitable Machine Learning in Healthcare Using User Parameters and Requirements, *Healthcare Analytics*, 100117, ISSN 2772-4425, <https://doi.org/10.1016/j.health.2022.100117>.
- [39] Mostafiz Ahammed, Md Al Mamun, Mohammad Shorif Uddin, A machine learning approach for skin disease detection and classification using image segmentation, *Healthcare Analytics* 2 (2022), 100122, <https://doi.org/10.1016/j.health.2022.100122>. ISSN 2772-4425.
- [40] Victor Chang, Meghana Ashok Ganatra, Karl Hall, Lewis Golightly, Qianwen Ariel Xu, An assessment of machine learning models and algorithms for early prediction and diagnosis of diabetes using health indicators, *Healthcare Analytics* 2 (2022), 100118, <https://doi.org/10.1016/j.health.2022.100118>. ISSN 2772-4425.
- [41] S. Arooj, Atta-ur-Rahman, M. Zubair, M.F. Khan, K. Alissa, M.A. Khan, A. Mosavi, Breast cancer detection and classification empowered with transfer learning, *Front. Public Health* 10 (2022), 924432, <https://doi.org/10.3389/fpubh.2022.924432>.
- [42] H. Rehman, N. Chandra, S. Rao Jammalamadaka, Competing risks survival data under middle censoring—an application to COVID-19 pandemic, *Healthcare Analytics* 1 (2021), 100006, <https://doi.org/10.1016/j.health.2021.100006>. ISSN 2772-4425.
- [43] UCI Machine Learning Repository'. <https://archive.ics.uci.edu/ml/index.php>.
- [44] P.D. Deba, H.K. Mahes Kumar, C. Chinmay, K. Mohammad, Review of machine and deep learning techniques in epileptic seizure detection using physiological signals and sentiment analysis, *ACM Transactions on Asian and Low-Resource Language Information Processing (TALLIP)* 1–29 (2022), <https://doi.org/10.1145/3552512>.
- [45] K. Rezaee, H. Ghayoumi Zadeh, C. Chakraborty, M. R. Khosravi, G. Jeon, "Smart visual sensing for overcrowding in COVID-19 infected cities using modified deep transfer learning," in *IEEE Transactions on Industrial Informatics*, vol. 19(1), 813–820, doi: 10.1109/TII.2022.3174160.
- [46] K. Rezaee, H. Ghayoumi Zadeh, C. Chakraborty, M. R. Khosravi, G. Jeon, "Smart visual sensing for overcrowding in COVID-19 infected cities using modified deep transfer learning," in *IEEE Transactions on Industrial Informatics*, vol. 19(1), 813–820, doi: 10.1109/TII.2022.3174160.
- [47] J. Arun Pandian, K. Kanchanadevi, D. Kumar, O. Geman, Deep convolutional generative adversarial network for metastatic tissue diagnosis in lymph node section, in: G.R. Kanagachidambaresan, D. Bhatia, D. Kumar, A. Mishra (Eds.), *System Design for Epidemics Using Machine Learning and Deep Learning. Signals and Communication Technology*, Springer, Cham, 2023, https://doi.org/10.1007/978-3-031-19752-9_9.
- [48] M.S. Devi, V.D. Kumar, A. Brezilianu, O. Geman, M. Arif, A novel blunge calibration intelligent feature classification model for the prediction of hypothyroid disease, *Sensors* 23 (2023) 1128, <https://doi.org/10.3390/s23031128>.
- [49] M.L. Zabara, I. Popescu, A. Burlacu, O. Geman, R.A.C. Dabija, I.V. Popa, C. Lupascu, Machine learning model validated to predict outcomes of liver transplantation recipients with hepatitis C: the Romanian national transplant agency cohort experience, *Sensors* 23 (2023) 2149, <https://doi.org/10.3390/s23042149>.